

# MECH5200 Numerical Methods for PDEs : *Video 26:* *Introduction to Discontinuous Galerkin Methods* *(DG-FEM)*

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References: These slides follow the presentation of Jan S. Hesthaven and Tim Warburton in, Nodal Discontinuous Galerkin Methods as well as the video introductory series by J. Bevan ([https://youtube.com/playlist?list=PLc4eVvgECLtptmgobxxXmVikejqPf4f8Dsi=7FrKxuMZ\\_dO1TH](https://youtube.com/playlist?list=PLc4eVvgECLtptmgobxxXmVikejqPf4f8Dsi=7FrKxuMZ_dO1TH))

# 1D Scalar Advection Equation

- In *Finite Volumes* (and to a lesser extent, *Finite Differences*) we discretized and solved the following **strong form** equation:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \quad (1)$$

- Where the flux,  $f(u) = a \cdot u$ .
- Initial conditions are defined such that:

$$u(x, 0) = u_0(x) \quad (2)$$

- And boundary conditions are defined based on the direction of wave propagation:

$$\begin{aligned} u(x_L, t) &= U_L \quad \text{if } a \geq 0 \\ u(x_R, t) &= U_R \quad \text{if } a \leq 0 \end{aligned}$$

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# DG-FEM – Solution Representation and Domain Discretization

- Let's start by defining our domain  $\Omega$  using  $K$  elements, each with independent nodes (see next slide for an example).
- Let's define the solution to the PDE at a location  $x$  in the domain as the (direct) summation of (non-overlapping) local element solutions:

$$u(x, t) \approx u_h(x, t) = \bigoplus_{k=1}^K u_h^k(x, t) \quad (3)$$

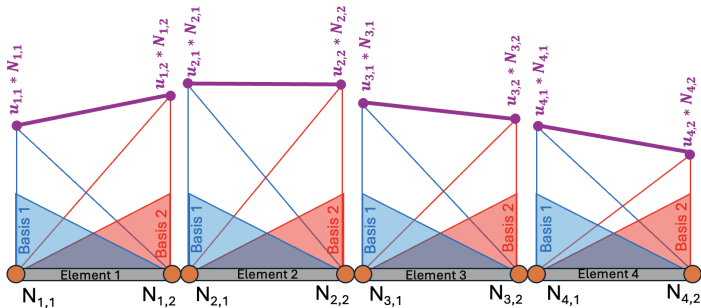
- Where **on each element**,  $k$ , a weighted Lagrange Polynomial basis is used to represent the solution:

$$u_h^k(x, t) = \sum_{n=1}^{N_p} \hat{u}_h^k(x_n, t) \cdot L_n^k(x) \quad (4)$$

- $L_n^k(x)$  is an Lagrange polynomial of order  $N_p - 1$

# DG-FEM – Solution Representation and Domain Discretization

- For a piecewise linear approximation of the solution this looks like:



- The solution (purple lines) is discontinuous between elements
- At the interface between elements, there are 2 nodes (one associated with element  $k$  and the other with element  $k + 1$ ).

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# DG-FEM – Developing the weak form of the equations

- Let's define the residual for a numerical approximation of the solution on an **element**:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = R(x, t) \quad (5)$$

- Next, define a weighting/testing basis  $w_i$  so that:

$$\int_{x_L}^{x_R} R_h(x, t) \cdot w_i(x) dx = 0 \quad (6)$$

- Consider a single element and multiply the strong form of the equation by the weighting/test function  $w_i$  and integrate over the element,  $D^k$ :

$$\int_{D^k} w_i \frac{\partial u_h}{\partial t} dx + \int_{D^k} w_i \frac{\partial (a \cdot u_h)}{\partial x} dx = \int_{D^k} w_i R(x) = 0 \quad (7)$$



# DG-FEM – Developing the weak form of the equations

- Now let's integrate the spatial term (second integral) by parts:

$$\int_{D^k} w_i \frac{\partial u_h}{\partial t} dx + \int_{D^k} a \cdot u_h^k \frac{\partial w_i}{\partial x} dx + \left[ a u_h^k w_i \right]_{x_k^L}^{x_k^R} = \int_{D^k} w_i R(x) dx = 0$$

- Introducing the element nodal basis function representations:

$$\begin{aligned} 0 &= \int_{D^k} w_i \frac{\partial \left( \sum_{n=1}^{N_p} \hat{u}_h^k(x_n, t) \cdot L_n^k(x) \right)}{\partial t} dx \\ &+ \int_{D^k} a \cdot \left( \sum_{n=1}^{N_p} \hat{u}_h^k(x_n, t) \cdot L_n^k(x) \right) \frac{\partial w_i}{\partial x} dx \\ &+ \left[ a u_h^k w_i \right]_{x_k^L}^{x_k^R} \\ &= 0 \end{aligned}$$

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# DG-FEM: Developing the Discrete Equations

- Which can be re-arranged and simplified (by dropping some sub/superscripts) as:

$$\sum \frac{\partial \hat{u}_j}{\partial t} \int_{D^k} w_i \mathbf{L}_j dx + \sum a \cdot \hat{u}_j \int_{D^k} \mathbf{L}_j^k \frac{\partial w_i}{\partial x} dx + \left[ au_h^k w_i \right]_{x_k^L}^{x_k^R} = 0$$

- For now, let's select basis functions on each element generically as,  $\phi_j$ , and  $\phi_i$  as we did in FEM (basis functions are Lagrange polynomials **local to the element**):

$$\sum \frac{\partial \hat{u}_j}{\partial t} \int_{D^k} \phi_i \phi_j dx + \sum a \cdot \hat{u}_j \int_{D^k} \phi_j \frac{\partial \phi_i}{\partial x} dx + [(au)\phi_i]_{x_k^L}^{x_k^R} = 0$$

- The unknown solution in this equation is only “connected to” other elements through the fluxes  $[(au)\phi_i]_{x_k^L}^{x_k^R}$ . (Different from FEM where basis functions connect elements).

# DG-FEM: Developing the Discrete Equations

- Which will take the following element-wise matrix form for the  $k - th$  element:

$$\underbrace{[M]_k}_{\text{Mass matrix}} \left[ \frac{d\hat{u}}{dt} \right]_k - a \cdot \underbrace{[S]_k}_{\text{stiffness mat}} [\hat{u}]_k = \underbrace{(a\hat{u}_k\phi_{i_{\text{left}}} - a\hat{u}_k\phi_{i_{\text{right}}})}_{\text{flux into and out of element}}$$

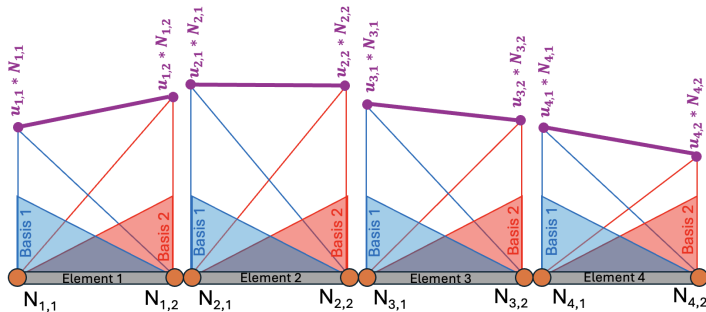
- Note: This looks a lot like a mix between finite volumes (the fluxes balancing with the time and spatial changes) and the finite element method (weak form of the equation satisfied using Galerkin method on each element).
- See next batch of slides for the integrals  $\rightarrow$  element matrices.

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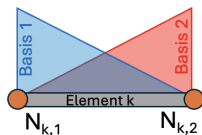
# DG-FEM: Developing the Element Matrices - Mass

- Recall our example geometry – let's work with linear basis functions for this example (but note that this is “easily” generalized to higher order):



# DG-FEM: Developing the Element Matrices - Mass

- Consider the interaction of the two basis functions on the element:



$$\phi_1 = \frac{x - x_2}{x_1 - x_2} = \frac{x - x_2}{-h}$$

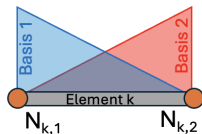
$$\phi_2 = \frac{x - x_1}{x_2 - x_1} = \frac{x - x_1}{h}$$

- Let's do an example integration here for the interaction of **Basis 1** and **Basis 1**:

$$\begin{aligned} I_1 &= \int_{D_K} \phi_1 \cdot \phi_1 \cdot dx = \int_{D_K} \frac{(x - x_2)(x - x_2)}{h^2} dx = \int_{D_K} \frac{x^2 - 2x_2x + x_2^2}{h^2} dx \\ &= \frac{1}{h^2} \left[ \frac{x^3}{3} - 2x_2 \frac{x^2}{2} + x_2^2 x \right]_{x=x_1}^{x=x_2} \rightarrow \frac{1}{h^2} \left[ \frac{x^3}{3} - 2h \frac{x^2}{2} + h^2 x \right]_{x=0}^{x=h} \\ &= \frac{1}{h^2} \left[ \frac{h^3}{3} - \cancel{2h \frac{h^2}{2}} + \cancel{h^2 h} \right]_{x=0}^{x=h} = \boxed{\frac{h}{3}} \end{aligned}$$

# DG-FEM: Developing the Element Matrices - Mass

- Consider the interaction of the two basis functions on the element:



$$\phi_1 = \frac{x - x_2}{x_1 - x_2} = \frac{x - x_2}{-h}$$

$$\phi_2 = \frac{x - x_1}{x_2 - x_1} = \frac{x - x_1}{h}$$

- Let's do the integration for the interaction of **Basis 1** and **Basis 2**:

$$\begin{aligned} I_1 &= \int_{D_K} \phi_1 \cdot \phi_2 \cdot dx = \int_{D_K} \frac{(x - x_2)(x - x_1)}{-h^2} dx = \int_{D_K} \frac{x^2 - x_2x - x_1x + x_2x_1}{-h^2} dx \\ &= -\frac{1}{h^2} \left[ x^3/3 - x_2x^2/2 - x_1x^2/2 + x_2x_1x \right]_{x=x_1}^{x=x_2} \rightarrow -\frac{1}{h^2} \left[ x^3/3 - hx^2/2 \right]_{x=0}^{x=h} \\ &= -\frac{1}{h^2} \left[ h^3/3 - h^3/2 \right]_{x=0}^{x=h} = \boxed{\frac{h}{6}} \end{aligned}$$



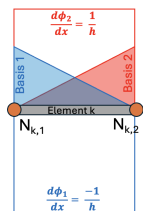
# DG-FEM: Developing the Element Matrices - Mass

- Evaluating all basis interactions on the element, the 1<sup>st</sup> integral  $\rightarrow$  element-based **mass** matrix:

$$[M] \left[ \frac{du}{dt} \right] = \int_{D_k} \phi_i \cdot \phi_j \cdot dx = \begin{pmatrix} \frac{h}{3} & \frac{h}{6} \\ \frac{h}{6} & \frac{h}{3} \end{pmatrix} \begin{pmatrix} \frac{du_{k,1}}{dt} \\ \frac{du_{k,2}}{dt} \end{pmatrix} \quad (8)$$

# DG-FEM: Developing the Element Matrices - Stiffness

- Consider the interaction of the two basis functions on the element for the second integral term:



$$\phi_1 = \frac{x - x_2}{x_1 - x_2} = \frac{x - x_2}{-h} \rightarrow \frac{d\phi_1}{dx} = -\frac{1}{h}$$

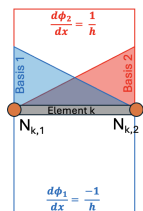
$$\phi_2 = \frac{x - x_1}{x_2 - x_1} = \frac{x - x_1}{h} \rightarrow \frac{d\phi_2}{dx} = \frac{1}{h}$$

- Let's do the integration for the interaction of **Basis 1** or **Basis 2** and **gradient** of **Basis 1**:

$$\begin{aligned} l_1 &= \int_{D_K} \phi_1 \cdot \frac{\partial \phi_1}{\partial x} \cdot dx = \int_{D_K} \frac{(x - x_2)(-1)}{-h^2} dx = \int_{D_K} \frac{(x - x_2)}{h^2} dx \\ &= \frac{1}{h^2} \left[ (x^2/2 - x_2 x) \right]_{x=x_1}^{x=x_2} \rightarrow \frac{1}{h^2} \left[ x^2/2 - hx \right]_{x=0}^{x=h} \\ &= \frac{1}{h^2} \left[ h^2/2 - h^2 \right]_{x=0}^{x=h} = \boxed{\frac{-1}{2}} \end{aligned}$$

# DG-FEM: Developing the Element Matrices - Stiffness

- Consider the interaction of the two basis functions on the element for the second integral term:



$$\phi_1 = \frac{x - x_2}{x_1 - x_2} = \frac{x - x_2}{-h} \rightarrow \frac{d\phi_1}{dx} = -\frac{1}{h}$$

$$\phi_2 = \frac{x - x_1}{x_2 - x_1} = \frac{x - x_1}{h} \rightarrow \frac{d\phi_2}{dx} = \frac{1}{h}$$

- Let's do the integration for the interaction of **Basis 1** or **Basis 2** and **gradient** of **Basis 2**:

$$\begin{aligned} l_1 &= \int_{D_K} \phi_1 \cdot \frac{\partial \phi_2}{\partial x} \cdot dx = \int_{D_K} \frac{(x - x_2)(1)}{-h^2} dx = - \int_{D_K} \frac{(x - x_2)}{h^2} dx \\ &= -\frac{1}{h^2} \left[ (x^2/2 - x_2 x) \right]_{x=x_1}^{x=x_2} \rightarrow -\frac{1}{h^2} [x^2/2 - hx]_{x=0}^{x=h} \\ &= -\frac{1}{h^2} [h^2/2 - h^2]_{x=0}^{x=h} = \boxed{\frac{1}{2}} \end{aligned}$$

# DG-FEM: Developing the Element Matrices - Stiffness

- The 2<sup>nd</sup> integral produces an element based **stiffness matrix**:

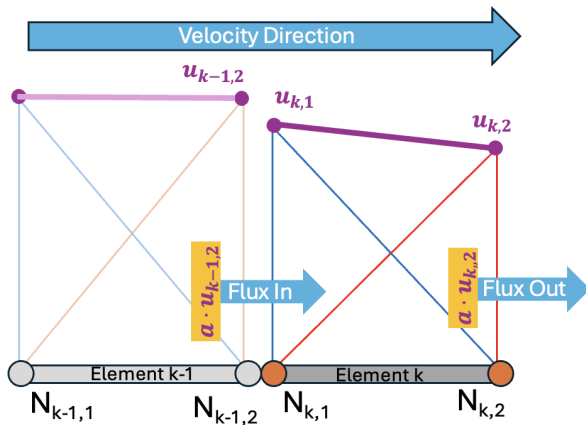
$$[S][u] = \int_{\Omega} \phi_j \frac{\partial \phi_i}{\partial x} dx = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} u_{k,1} \\ u_{k,2} \end{pmatrix} \quad (9)$$

- It is convenient to also include the unknown  $u_{k-1,2}$  associated with the node on the *upwind element* (we'll see why in a second):

$$[S][u] = \int_{\Omega} \phi_j \frac{\partial \phi_i}{\partial x} dx = \begin{pmatrix} 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} u_{k-1,2} \\ u_{k,1} \\ u_{k,2} \end{pmatrix} \quad (10)$$

# DG-FEM: Developing the Element Matrices – Fluxes

- Recall, for the upwind flux scheme, we assigned the upwind cell flux to the interface flux value. For each element:



# DG-FEM: Developing the Element Matrices – Fluxes

- When using upwinding, the flux on the left side of the  $k - th$  element will be:

$$a\hat{u}_k\phi_{i_{left}} = a \cdot \hat{u}_{k-1,2} \quad (11)$$

- And the flux on the right hand side of the  $k - th$  element will be:

$$a\hat{u}_k\phi_{i_{right}} = a \cdot \hat{u}_{k,2} \quad (12)$$

- The **upwind scheme** can now be written as a matrix.

$$(a\hat{u}_k\phi_{i_{left}} - a\hat{u}_k\phi_{i_{right}}) = a \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} u_{k-1,2} \\ u_{k,1} \\ u_{k,2} \end{pmatrix} \quad (13)$$

- Now, we see why we added the *upwind node* to the stiffness matrix (so it matches dimensions with the flux matrix here).

# DG-FEM: Developing the Element Matrices – Big Picture

- Hence, for each element, the following:

$$\underbrace{[M]_k}_{\text{Mass}} \left[ \frac{d\hat{u}}{dt} \right]_k - a \underbrace{[S]_k}_{\text{stiff}} [\hat{u}]_k = \underbrace{(a\hat{u}_k\phi_{i_{\text{left}}} - a\hat{u}_k\phi_{i_{\text{right}}})}_{\text{flux in and out of element}}$$

- Can be written as:

$$\begin{pmatrix} \frac{du_{k,1}}{dt} \\ \frac{du_{k,2}}{dt} \end{pmatrix} = a \cdot \underbrace{\begin{bmatrix} \frac{h}{3} & \frac{h}{6} \\ \frac{h}{6} & \frac{h}{3} \end{bmatrix}^{-1} \left( \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \right)}_{\text{Element matrix} \rightarrow \text{stamped} \rightarrow A_{\text{Global}}} \begin{bmatrix} u_{k-1,2} \\ u_{k,1} \\ u_{k,2} \end{bmatrix}$$

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# Implementation: Coding DG In MATLAB

- Try programming this now!
- Start with a pseudo code
- Perform elementwise setup
- Stamp/assemble element matrix into global matrix.
- Determine the approach for time discretization.
- Careful with the periodic boundary conditions.

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# DG-FEM: Examples

- <http://persson.berkeley.edu/movies.html>
- Here is another intro to DG:
- <http://persson.berkeley.edu/math228b/html/dgschool1.html>
- <https://www.joshbevan.com/media/test/>

# DG-FEM: Where to Learn More

- There's a lot of research in this area (especially for high-order methods)
- Good book: J.S.Hesthaven and T.Warburton, Nodal Discontinuous Galerkin Methods.
- Per-Olof Persson, Jaime Peraire, Ngoc Cuong Nguyen
- Bernardo Cockburn and Chi-Wang Shu
- K. Fidkowski, D. Darmofal
- And MANY others